Inverting the Schur complement, and large-dimensional Gelfand-Tsetlin patterns

Terence Tao · 16 September 2017

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Suppose we have an $n \times n$ matrix M that is expressed in block-matrix form as

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

where A is an $(n-k) \times (n-k)$ matrix, B is an $(n-k) \times k$ matrix, C is an $k \times (n-k)$ matrix, and D is a $k \times k$ matrix for some $1 < k < n$. If A is invertible, we can use the technique of Schur complementation to express the inverse of M (if it exists) in terms of the inverse of A, and the other components B, C, D of course. Indeed, to solve the equation

$$M(x \;\; y) = (a \;\; b),$$

where x, a are $(n-k) \times 1$ column vectors and y, b are $k \times 1$ column vectors, we can expand this out as a system

$$Ax + By = a$$

$$Cx + Dy = b.$$

Using the invertibility of A, we can write the first equation as

$$x = A^{-1}a - A^{-1}By \tag{1}$$

and substituting this into the second equation yields

$$(D - CA^{-1}B)y = b - CA^{-1}a$$

and thus (assuming that $D - CA^{-1}B$ is invertible)

$$y = -(D - CA^{-1}B)^{-1}CA^{-1}a + (D - CA^{-1}B)^{-1}b$$

and then inserting this back into (1) gives

$$x = (A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1})a - A^{-1}B(D - CA^{-1}B)^{-1}b.$$

Comparing this with

$$(x \;\; y) = M^{-1}(a \;\; b),$$

we have managed to express the inverse of M as

$$M^{-1} = \begin{pmatrix} A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D - CA^{-1}B)^{-1} \\ -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{pmatrix}. \tag{2}$$

One can consider the inverse problem: given the inverse M^{-1} of M, does one have a nice formula for the inverse A^{-1} of the minor A? Trying to recover this directly from (2) looks somewhat messy. However, one can proceed as follows. Let U denote the $n \times k$ matrix

$$U := \begin{pmatrix} 0 \\ I_k \end{pmatrix}$$

(with I_k the $k \times k$ identity matrix), and let V be its transpose:

$$V := (0 \;\; I_k).$$

Then for any scalar t (which we identify with t times the identity matrix), one has

$$M + UtV = \begin{pmatrix} A & B \\ C & D+t \end{pmatrix},$$

and hence by (2)

$$(M+UtV)^{-1} =$$

$$\begin{pmatrix} A^{-1} + A^{-1}B(D+t-CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D+t-CA^{-1}B)^{-1} \\ -(D+t-CA^{-1}B)^{-1}CA^{-1} & (D+t-CA^{-1}B)^{-1} \end{pmatrix}.$$

noting that the inverses here will exist for t large enough. Taking limits as $t \to \infty$, we conclude that

$$\lim_{t\to\infty} (M+UtV)^{-1} = \begin{pmatrix} A^{-1} & 0 \\ 0 & 0 \end{pmatrix}.$$

On the other hand, by the Woodbury matrix identity (discussed in this previous blog post), we have

$$(M+UtV)^{-1} = M^{-1} - M^{-1}U(t^{-1} + VM^{-1}U)^{-1}VM^{-1}$$

and hence on taking limits and comparing with the preceding identity, one has

$$\begin{pmatrix} A^{-1} & 0 \\ 0 & 0 \end{pmatrix} = M^{-1} - M^{-1}U(VM^{-1}U)^{-1}VM^{-1}.$$

This achieves the aim of expressing the inverse A^{-1} of the minor in terms of the inverse of the full matrix. Taking traces and rearranging, we conclude in particular that

$$\mathrm{tr}A^{-1} = \mathrm{tr}M^{-1} - \mathrm{tr}(VM^{-2}U)(VM^{-1}U)^{-1}. \tag{3}$$

In the $k=1$ case, this can be simplified to

$$\mathrm{tr}A^{-1} = \mathrm{tr}M^{-1} - \frac{e_n^T M^{-2} e_n}{e_n^T M^{-1} e_n} \tag{4}$$

where e_n is the nth basis column vector.

We can apply this identity to understand how the spectrum of an $n \times n$ random matrix M relates to that of its top left $n-1 \times n-1$ minor A. Subtracting any complex multiple z of the identity from M (and hence from A), we can relate the Stieltjes transform $s_M(z) := \frac{1}{n}\mathrm{tr}(M-z)^{-1}$ of M with the Stieltjes transform $s_A(z) := \frac{1}{n-1}\mathrm{tr}(A-z)^{-1}$ of A:

$$s_A(z) = \frac{n}{n-1} s_M(z) - \frac{1}{n-1} \frac{e_n^T (M-z)^{-2} e_n}{e_n^T (M-z)^{-1} e_n} \tag{5}$$

At this point we begin to proceed informally. Assume for sake of argument that the random matrix M is Hermitian, with distribution that is invariant under conjugation by the unitary group $U(n)$; for instance, M could be drawn from the Gaussian Unitary Ensemble (GUE), or alternatively M could be of the form $M = UDU^*$ for some real diagonal matrix D and U a unitary matrix drawn randomly from $U(n)$ using Haar measure. To fix normalisations we will assume that the eigenvalues of M are typically of size $O(1)$. Then A is also Hermitian and $U(n)$-invariant. Furthermore, the law of $e_n^T(M-z)^{-1}e_n$ will be the same as the law of $u^*(M-z)^{-1}u$, where u is now drawn uniformly from the unit sphere (independently of M). Diagonalising M into eigenvalues λ_j and eigenvectors v_j, we have

$$u^*(M-z)^{-1}u = \sum_{j=1}^n \frac{|u^* v_j|^2}{\lambda_j - z}.$$

One can think of u as a random (complex) Gaussian vector, divided by the magnitude of that vector (which, by the Chernoff inequality, will concentrate to $\sqrt{n}$). Thus the coefficients $u^* v_j$ with respect to

the orthonormal basis $v_1, \ldots, v_j$ can be thought of as independent (complex) Gaussian vectors, divided by that magnitude. Using this and the Chernoff inequality again, we see (for z distance ~ 1 away from the real axis at least) that one has the concentration of measure

$$u^*(M-z)^{-1}u \approx \frac{1}{n}\sum_{j=1}^{n}\frac{1}{\lambda_j - z}$$

and thus

$$e_n^T(M-z)^{-1}e_n \approx \frac{1}{n}\mathrm{tr}(M-z)^{-1} = s_M(z)$$

(that is to say, the diagonal entries of $(M-z)^{-1}$ are roughly constant). Similarly we have

$$e_n^T(M-z)^{-2}e_n \approx \frac{1}{n}\mathrm{tr}(M-z)^{-2} = \frac{d}{dz}s_M(z).$$

Inserting this into (5) and discarding terms of size $O(1/n^2)$, we thus conclude the approximate relationship

$$s_A(z) \approx s_M(z) + \frac{1}{n}(s_M(z) - s_M(z)^{-1}\frac{d}{dz}s_M(z)).$$

This can be viewed as a difference equation for the Stieltjes transform of top left minors of M. Iterating this equation, and formally replacing the difference equation by a differential equation in the large n limit, we see that when n is large and $k \approx e^{-t}n$ for some $t \geq 0$, one expects the top left $k \times k$ minor A_k of M to have Stieltjes transform

$$s_{A_k}(z) \approx s(t,z) \tag{6}$$

where $s(t,z)$ solves the Burgers-type equation

$$\partial_t s(t,z) = s(t,z) - s(t,z)^{-1}\frac{d}{dz}s(t,z) \tag{7}$$

with initial data $s(0,z) = s_M(z)$.

> **Example 1** If M is a constant multiple $M = cI_n$ of the identity, then $s_M(z) = \frac{1}{c-z}$. One checks that $s(t,z) = \frac{1}{c-z}$ is a steady state solution to (7), which is unsurprising given that all minors of M are also c times the identity.

> **Example 2** If M is GUE normalised so that each entry has variance σ^2/n, then by the semi-circular law (see previous notes) one has $s_M(z) \approx \frac{-z+\sqrt{z^2-4\sigma^2}}{2\sigma^2} = -\frac{2}{z+\sqrt{z^2-4\sigma^2}}$ (using an appropriate branch of the square root). One can then verify the self-similar solution
>
> $$s(t,z) = \frac{-z+\sqrt{z^2-4\sigma^2 e^{-t}}}{2\sigma^2 e^{-t}} = -\frac{2}{z+\sqrt{z^2-4\sigma^2 e^{-t}}}$$
>
> to (7), which is consistent with the fact that a top $k \times k$ minor of M also has the law of GUE, with each entry having variance $\sigma^2/n \approx \sigma^2 e^{-t}/k$ when $k \approx e^{-t}n$.

One can justify the approximation (6) given a sufficiently good well-posedness theory for the equation (7). We will not do so here, but will note that (as with the classical inviscid Burgers equation) the equation can be solved exactly (formally, at least) by the method of characteristics. For any initial position z_0, we consider the characteristic flow $t \mapsto z(t)$ formed by solving the ODE

$$\frac{d}{dt}z(t) = s(t,z(t))^{-1} \tag{8}$$

with initial data $z(0) = z_0$, ignoring for this discussion the problems of existence and uniqueness. Then from the chain rule, the equation (7) implies that

$$\frac{d}{dt} s(t, z(t)) = s(t, z(t))$$

and thus $s(t, z(t)) = e^t s(0, z_0)$. Inserting this back into (8) we see that

$$z(t) = z_0 + s(0, z_0)^{-1}(1 - e^{-t})$$

and thus (7) may be solved implicitly via the equation

$$s(t, z_0 + s(0, z_0)^{-1}(1 - e^{-t})) = e^t s(0, z_0) \tag{9}$$

for all t and z_0.

Remark 3 In practice, the equation (9) may stop working when $z_0 + s(0, z_0)^{-1}(1 - e^{-t})$ crosses the real axis, as (7) does not necessarily hold in this region. It is a cute exercise (ultimately coming from the Cauchy-Schwarz inequality) to show that this crossing always happens, for instance if z_0 has positive imaginary part then $z_0 + s(0, z_0)^{-1}$ necessarily has negative or zero imaginary part.

Example 4 Suppose we have $s(0, z) = \frac{1}{c-z}$ as in Example 1. Then (9) becomes

$$s(t, z_0 + (c - z_0)(1 - e^{-t})) = \frac{e^t}{c - z_0}$$

for any t, z_0, which after making the change of variables $z = z_0 + (c - z_0)(1 - e^{-t}) = c - e^{-t}(c - z_0)$ becomes

$$s(t, z) = \frac{1}{c - z}$$

as in Example 1.

Example 5 Suppose we have

$$s(0, z) = \frac{-z + \sqrt{z^2 - 4\sigma^2}}{2\sigma^2} = -\frac{2}{z + \sqrt{z^2 - 4\sigma^2}}.$$

as in Example 2. Then (9) becomes

$$s(t, z_0 - \frac{z_0 + \sqrt{z_0^2 - 4\sigma^2}}{2}(1 - e^{-t})) = e^t \frac{-z_0 + \sqrt{z_0^2 - 4\sigma^2}}{2\sigma^2}.$$

If we write

$$\begin{aligned} z &:= z_0 - \frac{z_0 + \sqrt{z_0^2 - 4\sigma^2}}{2}(1 - e^{-t}) \\ &= \frac{(1 + e^{-t})z_0 - (1 - e^{-t})\sqrt{z_0^2 - 4\sigma^2}}{2} \end{aligned}$$

one can calculate that

$$z^2 - 4\sigma^2 e^{-t} = (\frac{(1 - e^{-t})z_0 - (1 + e^{-t})\sqrt{z_0^2 - 4\sigma^2}}{2})^2$$

and hence

$$\frac{-z + \sqrt{z^2 - 4\sigma^2 e^{-t}}}{2\sigma^2 e^{-t}} = e^t \frac{-z_0 + \sqrt{z_0^2 - 4\sigma^2}}{2\sigma^2}$$

which gives

$$s(t,z) = \frac{-z + \sqrt{z^2 - 4\sigma^2 e^{-t}}}{2\sigma^2 e^{-t}}. \tag{10}$$

One can recover the spectral measure μ from the Stieltjes transform $s(z)$ as the weak limit of $x \mapsto \frac{1}{\pi}\mathrm{Im} s(x+i\varepsilon)$ as $\varepsilon \to 0$; we write this informally as

$$d\mu(x) = \frac{1}{\pi}\mathrm{Im} s(x+i0^+)\, dx.$$

In this informal notation, we have for instance that

$$\delta_c(x) = \frac{1}{\pi}\mathrm{Im}\frac{1}{c-x-i0^+}\, dx$$

which can be interpreted as the fact that the Cauchy distributions $\frac{1}{\pi}\frac{\varepsilon}{(c-x)^2+\varepsilon^2}$ converge weakly to the Dirac mass at c as $\varepsilon \to 0$. Similarly, the spectral measure associated to (10) is the semicircular measure $\frac{1}{2\pi\sigma^2 e^{-t}}(4\sigma^2 e^{-t} - x^2)_+^{1/2}$.

If we let μ_t be the spectral measure associated to $s(t,\cdot)$, then the curve $e^{-t} \mapsto \mu_t$ from $(0,1]$ to the space of measures is the high-dimensional limit $n \to \infty$ of a Gelfand-Tsetlin pattern (discussed in this previous post), if the pattern is randomly generated amongst all matrices M with spectrum asymptotic to μ_0 as $n \to \infty$. For instance, if $\mu_0 = \delta_c$, then the curve is $\alpha \mapsto \delta_c$, corresponding to a pattern that is entirely filled with c's. If instead $\mu_0 = \frac{1}{2\pi\sigma^2}(4\sigma^2 - x^2)_+^{1/2}$ is a semicircular distribution, then the pattern is

$$\alpha \mapsto \frac{1}{2\pi\sigma^2\alpha}(4\sigma^2\alpha - x^2)_+^{1/2},$$

thus at height α from the top, the pattern is semicircular on the interval $[-2\sigma\sqrt{\alpha}, 2\sigma\sqrt{\alpha}]$. The interlacing property of Gelfand-Tsetlin patterns translates to the claim that $\alpha\mu_\alpha(-\infty,\lambda)$ (resp. $\alpha\mu_\alpha(\lambda,\infty)$) is non-decreasing (resp. non-increasing) in α for any fixed λ. In principle one should be able to establish these monotonicity claims directly from the PDE (7) or from the implicit solution (9), but it was not clear to me how to do so.

An interesting example of such a limiting Gelfand-Tsetlin pattern occurs when $\mu_0 = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$, which corresponds to M being $2P - I$, where P is an orthogonal projection to a random $n/2$-dimensional subspace of $\mathbf{C}^n$. Here we have

$$s(0,z) = \frac{1}{2}\frac{1}{-1-z} + \frac{1}{2}\frac{1}{1-z} = \frac{z}{1-z^2}$$

and so (9) in this case becomes

$$s(t, z_0 + \frac{1-z_0^2}{z_0}(1-e^{-t})) = \frac{e^t z_0}{1-z_0^2}$$

A tedious calculation then gives the solution

$$s(t,z) = \frac{(2e^{-t}-1)z + \sqrt{z^2 - 4e^{-t}(1-e^{-t})}}{2e^{-t}(1-z^2)}. \tag{11}$$

For $\alpha = e^{-t} > 1/2$, there are simple poles at $z = -1, +1$, and the associated measure is

$$\mu_\alpha = \frac{2\alpha-1}{2\alpha}\delta_{-1} + \frac{2\alpha-1}{2\alpha}\delta_1 + \frac{1}{2\pi\alpha(1-x^2)}(4\alpha(1-\alpha) - x^2)_+^{1/2}\, dx.$$

This reflects the interlacing property, which forces $\frac{2\alpha-1}{2\alpha}\alpha n$ of the αn eigenvalues of the $\alpha n \times \alpha n$ minor to be equal to -1 (resp. $+1$). For $\alpha = e^{-t} \leq 1/2$, the poles disappear and one just has

$$\mu_\alpha = \frac{1}{2\pi\alpha(1-x^2)}(4\alpha(1-\alpha) - x^2)_+^{1/2}\, dx.$$

For $\alpha = 1/2$, one has an inverse semicircle distribution

$$\mu_{1/2} = \frac{1}{\pi}(1-x^2)_+^{-1/2}.$$

There is presumably a direct geometric explanation of this fact (basically describing the singular values of the product of two random orthogonal projections to half-dimensional subspaces of $\mathbf{C}^n$), but I do not know of one off-hand.

The evolution of $s(t,z)$ can also be understood using the *R-transform* and *S-transform* from free probability. Formally, letlet $z(t,s)$ be the inverse of $s(t,z)$, thus

$$s(t, z(t,s)) = s$$

for all t, s, and then define the R-transform

$$R(t,s) := z(t,-s) - \frac{1}{s}.$$

The equation (9) may be rewritten as

$$z(t, e^t s) = z(0,s) + s^{-1}(1-e^{-t})$$

and hence

$$R(t, -e^t s) = R(0, -s)$$

or equivalently

$$R(t,s) = R(0, e^{-t}s). \tag{12}$$

See these previous notes for a discussion of free probability topics such as the R-transform.

> **Example 6** If $s(t,z) = \frac{1}{c-z}$ then the R transform is $R(t,s) = c$.

> **Example 7** If $s(t,z)$ is given by (10), then the R transform is
>
> $$R(t,s) = \sigma^2 e^{-t} s.$$

> **Example 8** If $s(t,z)$ is given by (11), then the R transform is
>
> $$R(t,s) = \frac{-1 + \sqrt{1 + 4s^2 e^{-2t}}}{2se^{-t}}.$$

This simple relationship (12) is essentially due to Nica and Speicher (thanks to Dima Shylakhtenko for this reference). It has the remarkable consequence that when $\alpha = 1/m$ is the reciprocal of a natural number m, then $\mu_{1/m}$ is the free arithmetic mean of m copies of μ, that is to say $\mu_{1/m}$ is the free convolution $\mu \boxplus \ldots \boxplus \mu$ of m copies of μ, pushed forward by the map $\lambda \to \lambda/m$. In terms of random matrices, this is asserting that the top $n/m \times n/m$ minor of a random matrix M has spectral measure approximately equal to that of an arithmetic mean $\frac{1}{m}(M_1 + \ldots + M_m)$ of m independent copies of M, so that the process of taking top left minors is in some sense a continuous analogue of the process of taking freely independent arithmetic means. There ought to be a geometric proof of this assertion, but I do not know of one. In the limit $m \to \infty$ (or $\alpha \to 0$), the R-transform becomes linear and the spectral measure becomes semicircular, which is of course consistent with the free central limit theorem.

In a similar vein, if one defines the function

$$\omega(t,z) := \alpha \int_{\mathbf{R}} \frac{zx}{1-zx}\, d\mu_\alpha(x) = e^{-t}(-1 - z^{-1} s(t, z^{-1}))$$

and inverts it to obtain a function $z(t,\omega)$ with

$$\omega(t, z(t,\omega)) = \omega$$

for all t, ω, then the *S-transform* $S(t,\omega)$ is defined by

$$S(t,\omega) := \frac{1+\omega}{\omega} z(t,\omega).$$

Writing

$$s(t,z) = -z^{-1}(1 + e^t \omega(t, z^{-1}))$$

for any t, z, we have

$$z_0 + s(0,z_0)^{-1}(1-e^{-t}) = z_0 \frac{\omega(0,z_0^{-1}) + e^{-t}}{\omega(0,z_0^{-1}) + 1}$$

and so (9) becomes

$$-z_0^{-1} \frac{\omega(0,z_0^{-1})+1}{\omega(0,z_0^{-1})+e^{-t}} (1 + e^t \omega(t, z_0^{-1} \frac{\omega(0,z_0^{-1})+1}{\omega(0,z_0^{-1})+e^{-t}}))$$
$$= -e^t z_0^{-1}(1 + \omega(0,z_0^{-1}))$$

which simplifies to

$$\omega(t, z_0^{-1} \frac{\omega(0,z_0^{-1})+1}{\omega(0,z_0^{-1})+e^{-t}})) = \omega(0,z_0^{-1});$$

replacing z_0 by $z(0,\omega)^{-1}$ we obtain

$$\omega(t, z(0,\omega) \frac{\omega+1}{\omega+e^{-t}}) = \omega$$

and thus

$$z(0,\omega) \frac{\omega+1}{\omega+e^{-t}} = z(t,\omega)$$

and hence

$$S(0,\omega) = \frac{\omega+e^{-t}}{\omega+1} S(t,\omega).$$

One can compute $\frac{\omega+e^{-t}}{\omega+1}$ to be the S-transform of the measure $(1-\alpha)\delta_0 + \alpha\delta_1$; from the link between S-transforms and free products (see e.g. these notes of Guionnet), we conclude that $(1-\alpha)\delta_0 + \alpha\mu_\alpha$ is the free product of μ_1 and $(1-\alpha)\delta_0 + \alpha\delta_1$. This is consistent with the random matrix theory interpretation, since $(1-\alpha)\delta_0 + \alpha\mu_\alpha$ is also the spectral measure of PMP, where P is the orthogonal projection to the span of the first αn basis elements, so in particular P has spectral measure $(1-\alpha)\delta_0 + \alpha\delta_1$. If M is unitarily invariant then (by a fundamental result of Voiculescu) it is asymptotically freely independent of P, so the spectral measure of $PMP = P^{1/2}MP^{1/2}$ is asymptotically the free product of that of M and of P.

Comments (7)

 **domotorp** · 16 September 2017

There's a typo after "and then inserting this back into (1) gives", a there should be a parenthesis around the expression multiplied by a. Also, the equation (2) is too wide and the "(2)" is not visible (for me).
[Corrected, thanks – T.]

 **Sergei Ofitserov** · 20 September 2017

Dear Terence Tao! I again participate in discussion by theme Szemeredi's theorem(previous post). P(1),P(2),P(3)-that directions of unfolding arithmetic progressions. More exactly,P(1) and P(2)-that unfolding of arithmetic progressions by columns up and down. Upper bound-9999, lower bound-1000(that bounds of Fourier analysis). And here P(3)-that already direction of unfolding arithmetic progressions by stitchs, wear out in infinity. P(3) for theorem1.6-that line of locking in infinity(fully possibly-closed elliptical curve). 1-1/10L-that no finished first page of my book. Thanks! Sergei.

 **JDM** · 26 September 2017

Your inversion formulas also appear in Wikipedia. While everything is a "matrix", my Representation Theory textbooks hardly write out the operators in this way. By drawing a column for each group element, as Cayley or Schur might have done. And lo and behold they were block diagonal !!
https://en.m.wikipedia.org/wiki/Invertible_matrix#Blockwise_inversion
https://en.m.wikipedia.org/wiki/Block_matrix

 **Anonymous** · 28 September 2017

A closing bracket is missing in the equation number "(2)".
[It's there, but cut off by the blog. -T]

 **burakcakmakblog** · 2 October 2017

Professor, as regards the discussion below Example (8), the formula by Nica and Speicher, i.e. $R_A(s) = R_M(\beta s)$ for $\beta = (n-k)/n$ can be easily shown as follows: From the multiplicative free convolution we have $S_A(z) = S_M(\beta z)$. Then, by invoking the general relationship between the R- and S- transforms which is $R(s)S(sR(s)) = 1 = S(z)R(zS(z))$ (see [Haagerup, Larsen, 99]) we get $R_A(s) = R_M(\beta s)$.

 **Free fractional convolution powers | What's new** · 7 September 2020

[...] initial condition (see for a derivation). This equation can be solved explicitly using the emph as free probability analogues of the classical probability concepts of differential entropy and [...]

 **Anonymous** · 20 May 2026

I remember the techniques I learned in my linear algebra course for finding the inverse of a block matrix: such as elementary row operations, Cramer's rule, and so forth. I find the Schur complement a bit hard to memorize — is it a more practical or useful technique in practice?